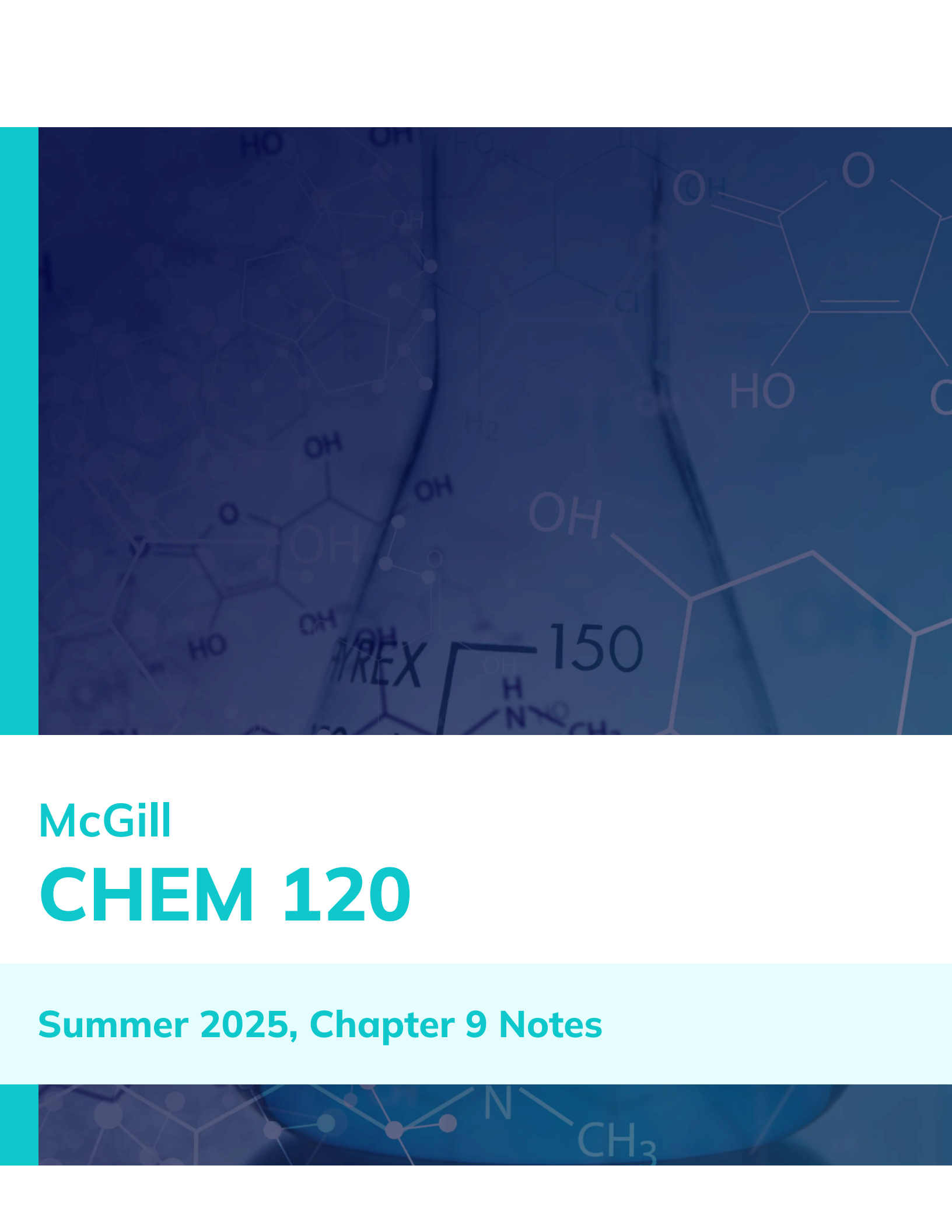




McGill CHEM 120

Summer 2025, Chapter 9 Notes

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9. Electrochemistry (Ch 16)

9.1 Oxidation States

9.1.1

Oxidation States

Oxidation state: A way to describe the degree of oxidation of a chemical

i WIZE TIP

Tips to Determine Oxidation State:

1. Free elements: oxidation state = 0

Examples: $\text{Cl}_2(\text{g})$ has an oxidation state of 0, $\text{Na}(\text{s})$ has an oxidation state of 0

2. Monoatomic ions: oxidation state = ionic charge

Examples: $\text{Cl}^-(\text{aq})$ has an oxidation state of -1, $\text{Na}^+(\text{aq})$ has an oxidation state of +1.

3. Hydrogen: oxidation state = +1

a. Exception: hydrogen in hydrides (e.g., NaH) – oxidation state = -1

4. Oxygen: oxidation state = -2

a. Exception: oxygen in peroxides (e.g., H_2O_2) – oxidation state = -1

5. For neutral molecules, oxidation numbers add up to zero.

For polyatomic ions, oxidation numbers add up to the ion charge.

Examples: $\text{Cl}_2(\text{g})$ has an oxidation state of 0, NaCl oxidation numbers add up to 0, SO_4^{2-} oxidation numbers add up to 2-.

Oxidation States - Additional Information

Other Low Yield Tips to Help Determine Oxidation States

- Alkali metals: oxidation state= +1
- Alkaline earth metals: oxidation state= +2
- Nitrogen Group: oxidation state= -3
- Oxygen family: oxidation state= -2
- Halogens: oxidation state= -1

Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓ Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra		104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Ms	116 Lv	117 Ts	118 Og
Lanthanides			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	
Actinides			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	

Note on Transition Metals

Transition metals can have different charges (ex. Pb^{2+} and Pb^{4+})
As a result, transition metals can also have different oxidation states!

Example:

What is the oxidation state of Cu in CuCl? _____

What is the oxidation state of copper in $\text{Cu}(\text{NO}_3)_2$? _____

When naming compounds involving transition metals, the **oxidation state of the transition metal is written in Roman numerals and put in brackets:**

- copper(I) chloride: CuCl
- copper(II) nitrate: $\text{Cu}(\text{NO}_3)_2$

Example: Oxidation State Practice

What is the oxidation state of:

S in S_2O_3 : _____

O in S_2O_3 : _____

What about Pb in Pb_2O_3 ?

This one is different because this compound is a mixed oxidation state compound, so although Pb should have 2+ or 4+, here, we get an oxidation state of _____ and that is ok because both Pb^{2+} and Pb^{4+} are present in the compound.

9.1.4

What is the oxidation state of Br in BrO_3^- ?

+4

☐

+5

☐

+6

☐

-4

☐

-5

☐

-6

☐

9.1.5

What is the oxidation state for the following: (enter answer as -1 or +2 for example)

a) P in P_2O_5 :

b) Cr in $\text{Na}_2\text{Cr}_2\text{O}_7$:

c) Al in $\text{Al}_2(\text{SO}_4)_3$:

d) Na in Na_2O_2 :

e) O in Na_2O_2 :

f) N in NH_3 :

9.2 Reduction-Oxidation (Redox) Reactions

9.2.1

Redox Reactions

OIL RIG! Oxidation Is Loss of electrons, Reduction Is Gain of electrons

OR

LEO GER! Loss of Electrons is Oxidation, Gain of Electrons is Reduction

OXIDATION (lose e^-)

REDUCTION (gain e^-)

Reducing agent reduces others and is itself oxidized.

Oxidizing agent oxidizes others and is itself reduced.

Oxidation number increases.

Oxidation number is reduced.

Keep in mind that in organic chemistry the definitions of oxidation and reduction are different:

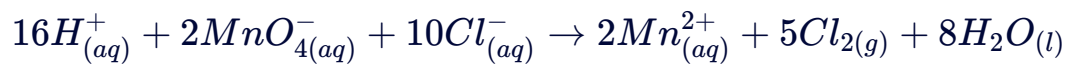
- For **reduction**, we would look for **more bonds to H** and **less bonds to O**
- For **oxidation**, we would look for **less bonds to H** and **more bonds to O!**

How can you recognize if a redox reaction is occurring?

1. Can **check oxidation #s of each element** to see if they **change from reactants to products** (most important for now!)
2. Familiarize yourself with **common oxidizing and reducing agents** (you'll learn more about this in orgo!)
 - **Oxidizing agents:** O_2 , halogens, HNO_3 , $\text{Cr}_2\text{O}_7^{2-}$, MnO_4^-
 - **Reducing agents:** H_2 , metals, C

Practice: Oxidation vs Reduction

Consider the following balanced redox equation:



Which species is being oxidized?

☐ H^{+}

☐ MnO_4^{-}

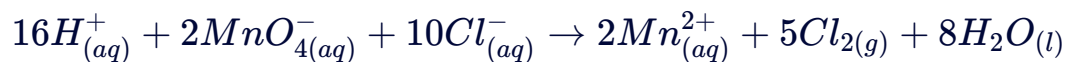
☐ Cl^{-}

☐ Mn^{2+}

☐ Cl_2

Practice: Oxidizing vs Reducing Agent

Consider the following balanced redox equation:



Which species is the reducing agent?

H^{+}

☐

MnO_4^{-}

☐

Cl^{-}

☐

Mn^{2+}

☐

Cl_2

☐

Practice: Oxidizing vs Reducing Agents

What is the reducing agent in the following chemical reaction?



Au^+

☐

NO

☐

NO_3^-

☐

Au

☐

9.3 Balancing Redox Reactions [half-reaction method]

9.3.1

Balancing Redox Reactions: Half-Reaction Method

The "half-reaction method" of balancing redox reactions is the most commonly used method :)

Balance the following redox reaction in basic conditions and assign oxidation numbers to all species.



WIZE TIP

Balancing redox equations:

1. **Check the oxidation numbers** of each species by inspection (**change in oxidation # means this is a redox reaction!**)
2. **Write** the reduction and oxidation **half-reactions**.
3. **Balance each** half-reaction: start with **atoms that are neither O nor H**, balance oxygens with H_2O , hydrogens with H^+ , and remaining charges with e^- .
4. **Multiply** the reduction and oxidation half-reactions as needed so that each reaction exchanges the **same number of e^-** .
5. **Add** the oxidation and reduction half reactions together and simplify.
6. **Additional step in basic solution only:** there can't be any H^+ ions, so **neutralize the H^+ by adding the same # of OH^- to each side**.

! WATCH OUT!

Always solve in acidic conditions first!

Then, if the question is asking for basic conditions do step #6!

1. Check the oxidation numbers of each species by inspection (change in oxidation # means this is a redox reaction!)

Here we are told it's a redox reaction already, but we can still check oxidation #s to see what is oxidized and what is reduced:



- Ag goes from having an oxidation # of _____ to _____. Therefore Ag is _____.
- Zn goes from having an oxidation # of _____ to _____. Therefore Zn is _____.

2. Write the reduction and oxidation half-reactions.

Oxidation Half Reaction:

Reduction Half Reaction:

3. Balance each half-reaction: start with atoms that are neither O nor H, balance oxygens with H_2O , hydrogens with H^+ , and remaining charges with e^- .

a) Balance everything that is not O and H:

ox:

red:

ox: $2\text{Ag(s)} \rightarrow \text{Ag}_2\text{O(aq)}$

red: $\text{Zn}^{2+}(\text{aq}) \rightarrow \text{Zn(s)}$

b) Balance O with H_2O :

ox:

red:

c) Balance H with H^+

ox:

red:

d) Balance charges with electrons!

ox:

red:

ox:

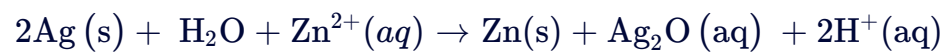


red:



4. Multiply the reduction and oxidation half-reactions as needed so that each reaction exchanges the same number of e⁻.

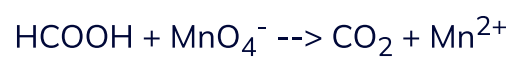
5. Add the oxidation and reduction half reactions together and simplify.



6. Additional step in basic solution only: there can't be any H⁺ ions, so neutralize the H⁺ by adding the same # of OH⁻ to each side.

Practice: Balancing Redox Reactions

Balance the following redox reaction in an acidic solution.



9.4 Reduction Potentials

9.4.1

Reduction Potentials

- Each half reaction has a potential.
- For example, the first half reaction below has a reduction potential of -3.05V !
 - We got these potentials from measuring the voltage when the other half reaction was the **reference electrode (H 0V)**
- In the table below, are the reactions showing reductions or oxidations? _____
 - Therefore, these are _____ potentials!
- **If we flip a reaction, the E° potential changes signs**, let's do this for the first half reaction in the table so it is clear:
 - Is this reaction now describing reduction or oxidation? _____!
 - So do you think the potential written would now be a reduction or oxidation potential? _____ potential
 - **In general:**
 - **If we see a more positive reduction potential:**
 - It means something is more likely to be oxidized/reduced: _____
 - and will act as a(n) reducing/oxidizing agent: _____ agent
 - **If we see a more negative reduction potential:**
 - It means something is more likely to be oxidized/reduced: _____
 - and will act as a(n) reducing/oxidizing agent: _____ agent

Half Reaction	Potential E°
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.05
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.36
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$	-1.67
$\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.83
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.44
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.23
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13
$\text{Fe}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.036
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	0
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	0.34
$\text{Cu}^+(\text{aq}) + \text{e}^- \rightarrow \text{Cu}(\text{s})$	0.52
$\text{Hg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Hg}(\text{l})$	0.8
$\text{Ag}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ag}(\text{s})$	0.8
$\text{Pt}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pt}(\text{s})$	1.2
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$	1.23
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s})$	1.5
$\text{F}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{F}^-(\text{aq})$	2.87

- Based on this table, answer the following questions:

1) What is the strongest oxidizing agent? _____

2) What is the weakest reducing agent? _____

3) What is the strongest reducing agent? _____

! WATCH OUT!

DO NOT assume you are given REDUCTION potentials in the table!

Here we know they are reduction potentials because we see electrons being added on the left sides of the equations.

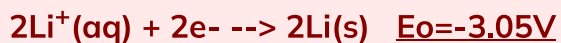
Your prof may try to trick you and show you oxidation potentials instead, where electrons would be seen on the right sides of the equations.

Another thing we have to know is that we may need to multiply half reactions by a coefficient (later on in problems). When we do this NEVER multiply the potential by a coefficient.

For example in the table we have:



If I needed to multiply this half reaction by 2, I would get:



How Potentials Relate to Spontaneity

- The overall cell potential will determine the spontaneity of redox reactions:

$$\Delta G^{\circ} = -nFE_{redox}^{\circ}$$

E_{redox}° = the potential associated with the redox reaction

n = # of electrons

F = Faraday's constant (96 500 C/mol e⁻)

ΔG° = Gibbs free energy

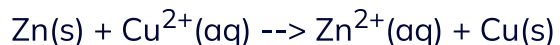
- According to this equation, **when E_{redox}° is positive**.... ΔG° is positive or negative:

- This means the reaction would be spontaneous/non-spontaneous: _____
 - When this is the case we have a **galvanic cell** (more on this soon!)
 - K _____ 1

- When E_{redox}° is negative**.... ΔG° is positive or negative: _____

- This means the reaction would be spontaneous/non-spontaneous: _____
 - When this is the case we have an **electrolytic cell** (more on this soon!)
 - K _____ 1

Apply: If we had the following overall reaction, would it be spontaneous or not?



1) Let's write out the oxidation and reduction half reactions:

a) Oxidation half reaction:

b) Reduction half reaction:

2) Given the above reactions that are occurring, write out an oxidation potential and a reduction potential using the table values below:

From the table:



For us, _____ is being oxidized, so $E^{\circ}_{\text{ox}} =$ _____ (changes sign)

And _____ is being reduced, so $E^{\circ}_{\text{red}} =$ _____ (stays the same)

3) Solve for E°_{redox} :

$$E^{\circ}_{\text{redox}} = E^{\circ}_{\text{ox}} + E^{\circ}_{\text{red}}$$

$$E^{\circ}_{\text{redox}} = 0.76 + 0.34$$

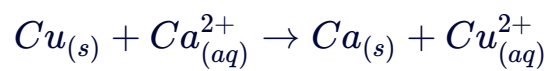
$$E^{\circ}_{\text{redox}} = 1.10\text{V}$$

4) Figure out if the reaction is spontaneous using this equation: $\Delta G^{\circ} = -nFE^{\circ}_{\text{redox}}$

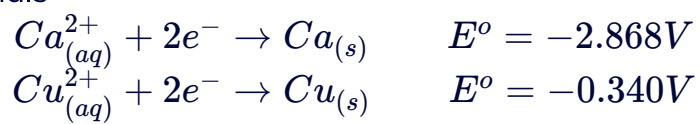
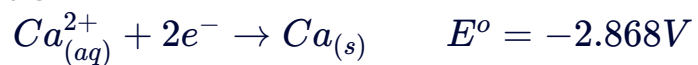
- Our E°_{redox} value is positive/negative: _____
- This means our ΔG° value is positive/negative: _____
- Therefore, the reaction is spontaneous/non-spontaneous: _____

9.4.3

Will the following reaction proceed spontaneously or will current need to be added externally?



Standard reduction potentials



9.4.4

Based on the following reduction potentials, which of these species is the strongest reducing agent?

Reaction	$E^\circ (V)$
$2HClO(aq) + 2H^+(aq) + 2e^- \rightarrow Cl_2(g) + 2H_2O(l)$	+1.63
$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$	+1.36
$Br_2(l) + 2e^- \rightarrow 2Br^-(aq)$	+1.08
$I_2(s) + 2e^- \rightarrow 2I^-(aq)$	+0.535
$Ag^+(aq) + e^- \rightarrow Ag(s)$	+0.799
$Sn^{4+}(aq) + 2e^- \rightarrow Sn^{2+}(aq)$	+0.15

☐ $I_2 (s)$ ☐ $I^- (aq)$ ☐ $Br_2 (l)$ ☐ $Br^- (aq)$ ☐ $Ag(s)$ ☐ $Sn^{2+} (aq)$ ☐ $Sn^{4+} (aq)$

9.5 Galvanic Cells

9.5.1

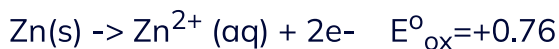
Galvanic/Voltaic Cells

- A galvanic cell constructed of two half cells is shown below.
- Before we take a deeper look, **let's review what we already know about galvanic cells:**
 - **Do galvanic cells have spontaneous or non-spontaneous electron flow?** _____
 - **Is the G° positive or negative?** _____
 - **According to the equation $\Delta G^\circ = -nFE^\circ_{\text{redox}}$ then is the E°_{redox} positive or negative?** _____

- **Let's take a look at our 2 half reactions from the table again:**

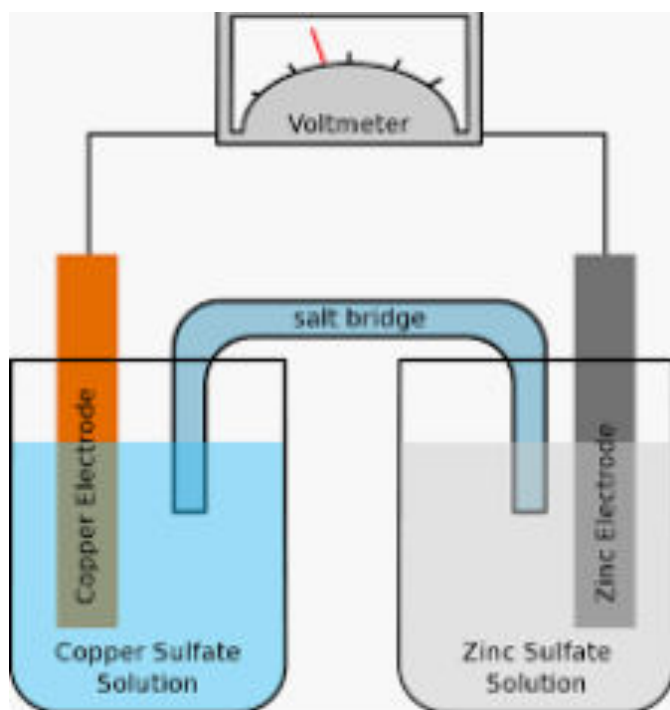


- **In order to get an E°_{redox} that is positive, we needed to flip the Zn(s) equation:**



$E^\circ_{\text{redox}} = 1.1\text{V}$ (you might see the E°_{cell} be referred to as the electromotive force or emf, they mean the same thing!)

Now let's label the galvanic cell...



- 1) Which is the reduction half equation and which is the oxidation half reaction?
- 2) Using the following "Wize Concept," label the anode and cathode in the above diagram.

 WIZE CONCEPT

The anode is always where oxidation happens.

The cathode is always where reduction happens.

There are different ways to remember this. One way is to remember that anode and oxidation both start with vowels, while cathode and reduction both start with consonants. The other way to remember this is: An Ox Red Cat! Pick your favourite way :)

3) Show where the ions are moving (into solution/onto electrode)

4) Which electrode is shrinking? Which is growing?

5) Consider the salt bridge. Let's say it's made of KCl. Which half cell would the K^+ ions travel to? Which half cell would the Cl^- ions travel to?

 WIZE CONCEPT

Anions of the salt bridge move to the anodic solution

Cations of the salt bridge move to the cathodic solution.

This helps to complete the circuit and prevent charges from building up by keeping the solution neutral.

6) Draw the direction of electron flow.

- Analogy: Running out of your last exam! Everyone is running in the same direction and to where they want to go (outside of the school!)

 WIZE TIP

Since the anode is always where oxidation happens and the cathode is always where reduction happens, then electrons will always flow from the anode to the cathode!

7) Write in the charges (+ or -) for each electrode.

 WIZE CONCEPT

Since galvanic cells involve spontaneous electron flow, electrons will travel to where they want to go!

Electrons will always go from the anode to the cathode.

But here, since it's spontaneous flow it means that the cathode will have a positive charge (electrons want to go here) and the anode will have a negative charge (electrons want to get away from here!)

We will see how this is different for electrolytic cells...

8) Write in the spectator ions (in this case it is SO_4^{2-} , you might also see NO_3^- in other cases).

- These ions are called "spectator" ions because that's all they are doing...spectating! They don't participate in the actual reaction, but help to balance charges!

9.5.2

In a galvanic cell, oxidation occurs at the:

anode

☐

cathode

☐

salt bridge

☐

electrolyte

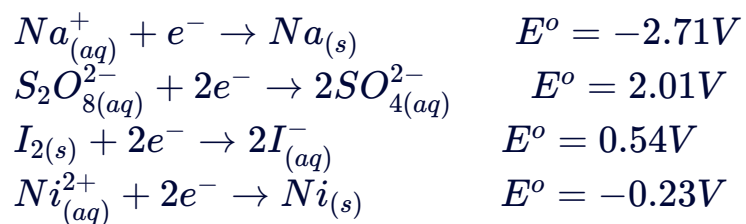
☐

in cathodic space

☐

9.5.3

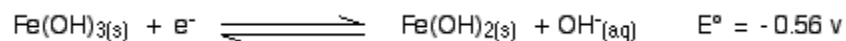
Construct a galvanic cell with the highest voltage given the following half reactions. Determine the voltage produced and express it in standard cell notation.



Will a Reaction Occur?

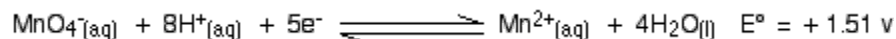
Will oxygen oxidise iron(II) hydroxide to iron(III) hydroxide under alkaline conditions?

The E° values are:



Will chlorine oxidise manganese(II) ions to manganate(VII) ions?

The E° values are:



Will Zn^{2+} react with $\text{Cu}(\text{s})$?



You are told that $\text{I}_2(\text{s})$ and $\text{Ag}(\text{s})$ do not react. Without using reduction potentials, what does this tell us about their relative reduction potentials?

Let's write out this reaction:



How could we have a reaction using the reactants and products in the above equation?



What is being oxidized and what is being reduced in our reaction?

oxidized: _____

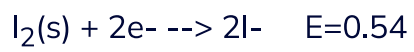
reduced: _____

What can we say about the reduction potentials?

_____ SRP > _____ SRP

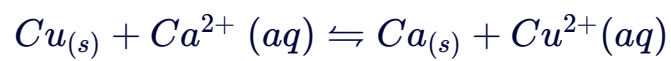
Let's check:

On SRP table it says:



9.5.5

Given the half cells below, which direction will the following reaction proceed?



9.5.6

Given the following half cells, what is the maximum voltage of a battery that could be constructed?



Answer

9.6 Cell Notation

9.6.1

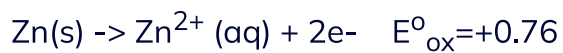
Cell Diagram

- The **cell diagram aka standard cell notation** is as follows:

| Anode | Ions from anode || Ions to cathode | Cathode |

If we wanted to write this notation for the galvanic cell we just drew...

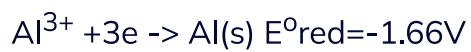
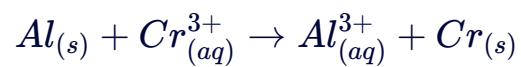
- These were the 2 half reactions that were involved:



- It can also be called **standard** cell notation because it is describing **SATP**: _____ C, _____ atm, [_____ M]
- Note that if a half cell does not contain a solid species then an inert electrode must be used. eg. $\text{Pt}_{(\text{s})}$

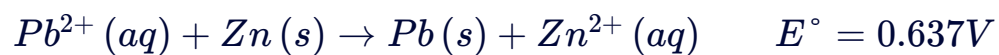
9.6.2

Write the cell notation for the voltaic (or galvanic) cell that incorporates the following redox reaction:



9.6.3

In a zinc-lead cell the reaction is:



Which of the following statements about this cell is FALSE?

The zinc electrode is the cathode

☐

The reaction will go in the direction indicated

☐

The shorthand notation is $Zn(s) \mid Zn^{2+}(aq) \parallel Pb^{2+}(aq) \mid Pb(s)$

☐

The lead electrode is positively charged

☐

9.7 Important Equations + Nonstandard conditions (Nernst Equation)

9.7.1

Cell Potentials and Gibbs Free Energy

- Free energy (ΔG°) and cell potential (E°_{cell}) are related through the following equation:

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

ΔG° = Free energy difference between products and reactants in their standard states

n = # moles of electrons

F = Faraday constant = $96485 \frac{C}{\text{mol } e^-}$

E°_{cell} = Standard electrochemical reduction potential

**** If E°_{cell} is positive, the process is spontaneous ****

- Remember that ΔG° and E° are for the cell at standard conditions (1 atm for gases and 1M for solutions)

$$\Delta G^\circ = -nFE^\circ_{\text{cell}} = -RT \ln K$$

$$E^\circ_{\text{cell}} = \frac{RT}{nF} \ln K$$

$$K = e^{\frac{nFE^\circ}{RT}}$$

- Under non-standard conditions we have:

$$\Delta G = \Delta G^\circ + RT \ln Q = -nFE_{\text{cell}}$$

- We can correct the E° to reflect the non-standard conditions by using the Nernst equation

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF} \ln Q \text{ or } E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.05916V}{n} \log Q$$

$$R = \text{gas constant} \left(\frac{8.314J}{\text{mol } K} \right)$$

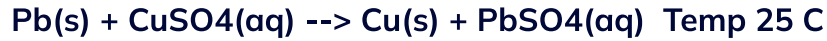
$$T = \text{temperature}$$

$n = \# \text{ of electrons}$

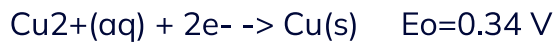
$F = \text{Faradays Constant}$

9.7.2

Calculate the ΔG° and K_{eq} for following electrochemical cells: $\Delta G^\circ = -nFE^\circ_{cell}$ & $K_{eq} = e^{nFE^\circ_{cell}/RT}$



1) From a standard reduction potential table we can find the following:



2) Let's try to solve for ΔG° first:

$$\Delta G^\circ = -nFE^\circ_{cell}$$

We know n = moles of electrons involved

Here, $n =$ _____

F is just a constant (Faraday's constant): 96500 C/mole-

E°_{cell} we can calculate by using the 2 half reactions, reversing a reaction based on the overall chemical reaction, and then adding up the E° values for oxidation and reduction:

In our overall equation, when we look at the reactants, which is being oxidized and which is being reduced?

Based on this info, which equation should we reverse from the table? _____

What is the E°_{cell} (redox)=? _____

Now we can plug these values in and solve for ΔG° :

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

If we wanted our ΔG° to be in kJ/mol what would our answer be? _____

3) Now let's solve for K_{eq} :

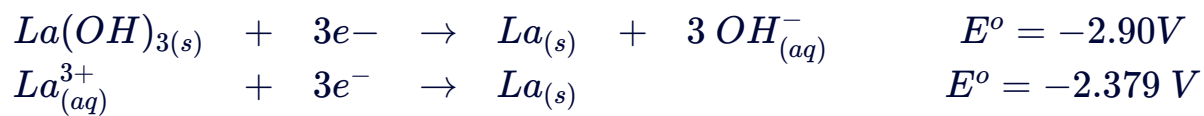
$$K_{\text{eq}} = e^{nFE^\circ/RT}$$

We already found n and E° , we know F (96500 C/mole-), we know R (8.314 J/molK) and the temperature is given in the question (25°C=298K!) Note: These equations are used for standard conditions, so the temperature would have to be 25°C or 298K!

$$K_{\text{eq}} = e^{nFE^\circ/RT}$$

9.7.3

Use the following reduction potentials to calculate the K_{sp} of $\text{La}(\text{OH})_3(s)$



3.64×10^{-27}

☐

3.56×10^{-27}

☐

3.67×10^{-27}

☐

3.54×10^{-27}

☐

Cells Operating at Non-Standard Conditions: The Nernst Equation

- If we want to operate a cell at non-standard conditions (not 1 M solutions) we need to use the Nernst equation to find new electromotive force

$$E_{cell} = E_{cell}^{\circ} - \frac{RT}{nF} \ln Q$$

R: The ideal gas constant (8.314 J/molK)

T: Temperature (K)

n: # of electrons involved

F: The Faraday Constant (96500 C/mol e-)

E°_{cell} : standard electromotive force

9.7.5

Determine the voltage produced in a galvanic cell at room temperature made up of the following half reactions at a pH of 3.5 and the following concentrations. $[\text{MnO}_4^-] = 0.5 \text{ M}$, $[\text{Mn}^{2+}] = 0.4 \text{ M}$, $[\text{Sn}^{4+}] = 2.0 \text{ M}$ and $[\text{Sn}^{2+}] = 0.8 \text{ M}$ $R=8.314\text{J/molK}$



Using the Nernst Equation:

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{nF} \ln Q$$

9.7.6

At 25C, what is the potential of the following cell and what is the K_{eq} ?



1) Is the reaction occurring in standard or non-standard conditions?

If it is occurring in standard conditions --> it is asking us to solve for E°_{cell}

If it is occurring in nonstandard conditions --> it is asking us to solve for E_{cell}

2) Which equation will we use to solve this problem?

Nernst equation for non-standard conditions!

$$E = E_o - (RT/nF)\ln(Q)$$

E is what the question is asking us to solve

E_o we can calculate by using the SRP from a table

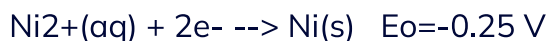
$R=8.314\text{J/molK}$

$n=\#$ of electrons involved in the overall reaction

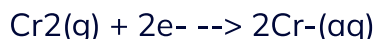
$F=\text{faraday's constant } (96500\text{C/mole-})$

$Q=[\text{products}]/[\text{reactants}]$

3) Solve for Ecell using SRP tables:



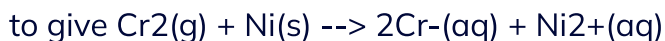
From this the standard cell reaction would be:



$E_o =$



$E_o =$



$E_o =$

4) Now we can determine E at non-standard conditions via the NERNST Equation:

$$E = E_o - (RT/nF)\ln(Q)$$

Bonus: Is this reaction spontaneous or not?

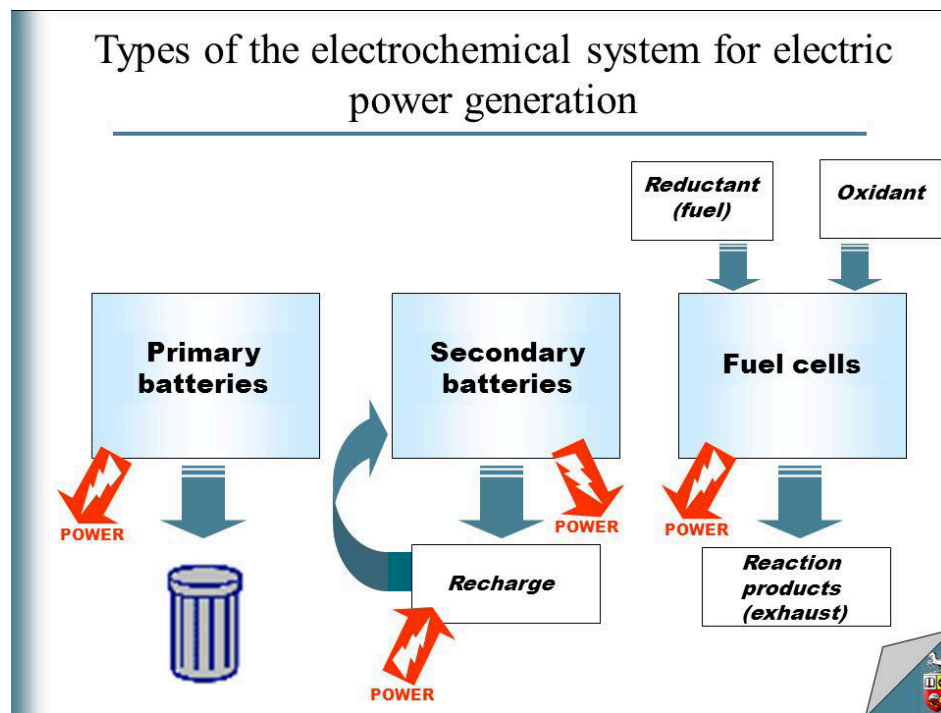
5) What is K_{eq} at 298 K?

$$K_{eq} = e^{nFE_o/RT}$$

9.8 Other Applications (including Batteries and Corrosion)

9.8.1

There are 3 main types of batteries:

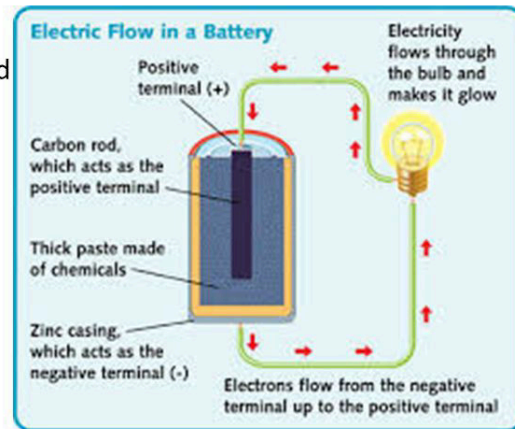


1) Primary batteries

- non-rechargeable (discarded when the cell reaction reaches equilibrium)
- one way (irreversible)
- self-contained series of voltaic cells
- ex. modern alkaline battery, silver button battery (like the one we see in watches), lithium battery,
- alkaline cells is about 1.5V

Primary Cell

- One use (non-rechargeable/disposable)
- Chemical reaction used, can not be reversed
- Used when long periods of storage are required
- Lower discharge rate than secondary batteries
- Use:
 - smoke detectors, flashlights, remote controls

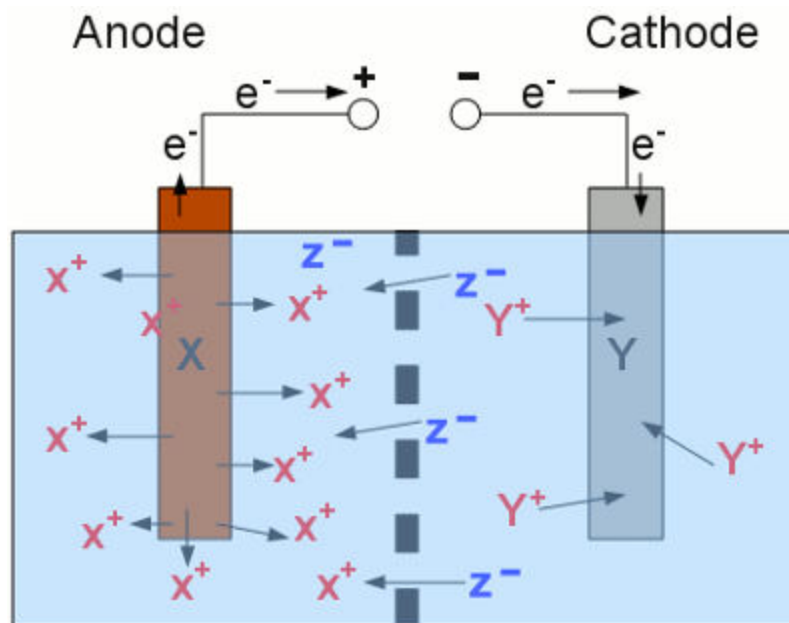


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2) Secondary Batteries

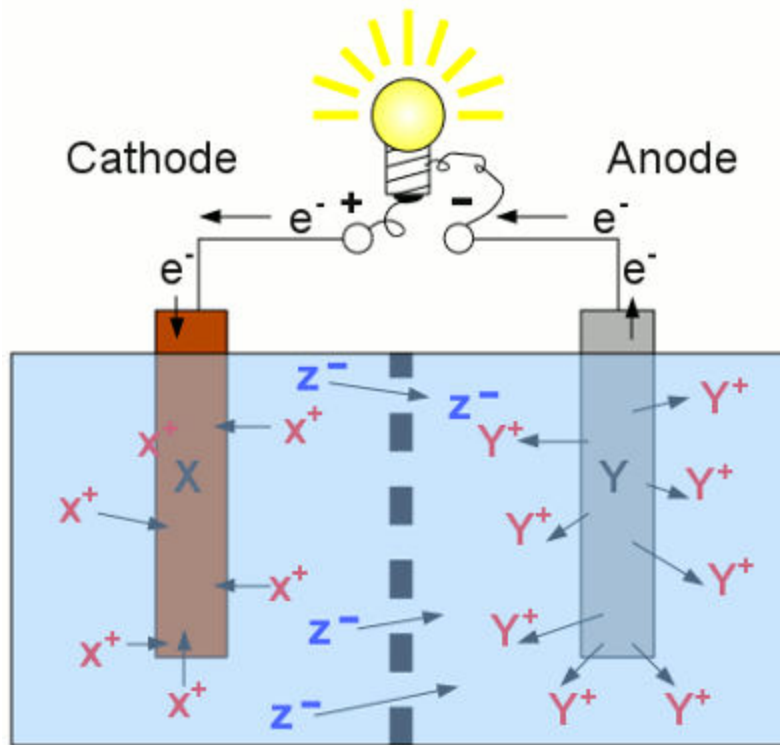
- rechargeable
- self-contained series of voltaic cells
- electrical current supplied to reverse the cell reaction
- ex: lead-acid batteries (car batteries), nickel-metal hydride batteries (power tools), lithium ion battery (laptops, cell phone)

a) Charging

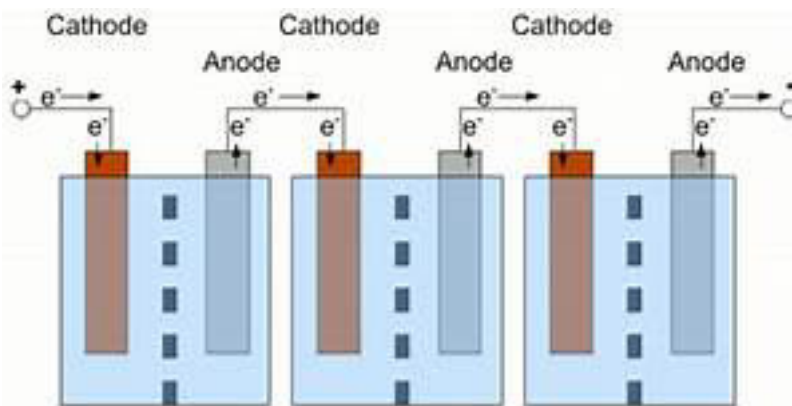


- electrolysis (not a voltaic cell)
- need a voltage source
- reaction stops when the cathodic solution runs out of cations

b) Discharging

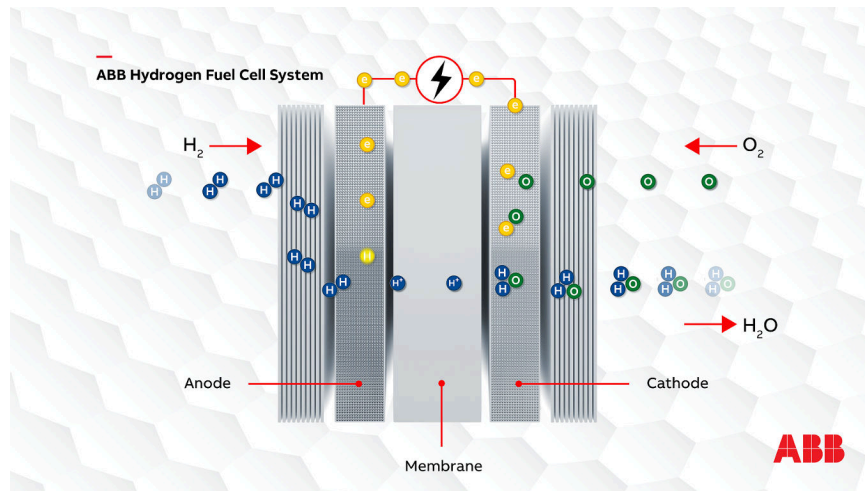


- this acts like a galvanic cell (spontaneous electron flow)
- reverse process is occurring, metal on right is now the anode, metal on left is the cathode



3) Fuel Cells

- non self-contained voltaic cells
- controlled combustion (O_2 and H_2 enter cells, H_2O leaves cells)
- can flow chemicals through system to generate electricity



Rechargeable Batteries

- We will soon see that rechargeable batteries can act as both galvanic and electrolytic cells!
- Think of our phone batteries!
 - If we unplug it from our charger in the morning, then from that point onwards the battery is **discharging** (we will soon see that this is a spontaneous process)



- Then at night, when we go to plug it into the charger, the battery is **charging** (this part requires a battery, which is an external source of energy so this part is non-spontaneous)

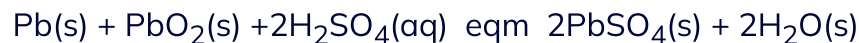


- Based on this info do you think the discharging process is more like a galvanic or electrolytic cell?

-
- What about for the charging process? _____

Let's consider a rechargeable lead-acid battery...

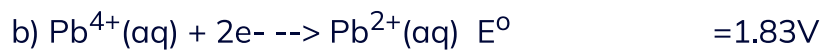
The overall equation for this battery is:



1) Draw in the oxidation states for Pb in the above equation

2) Now let's consider the discharge process

We have these 2 reactions:



$E^\circ_{\text{redox}} =$ _____

Is this process spontaneous or not?

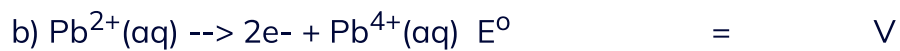
We can now draw out the cell for this: _____ cell

Make one electrode Pb(s) and the other PbO₂(s) (which is Pb⁴⁺)

- Is a battery needed for this reaction? _____
- label the anode and the cathode
- Show the direction in which electrons flow
- Label the charges of both the anode and the cathode

3) Now let's consider the charging process

To charge the battery, we have the same 2 reactions but want them to run in reverse:



$E^\circ_{\text{redox}} =$ _____ V

Is this process spontaneous or not?

We can now draw out the cell for this: _____ cell

Make one electrode PbSO_4 (which is also Pb^{2+}) and the other electrode the same.

- Is a battery needed for this reaction? _____
- Label the anode and the cathode
- Show the direction in which electrons flow
- Label the charges of both the anode and the cathode
- What are the charges on the battery (if there is one?)



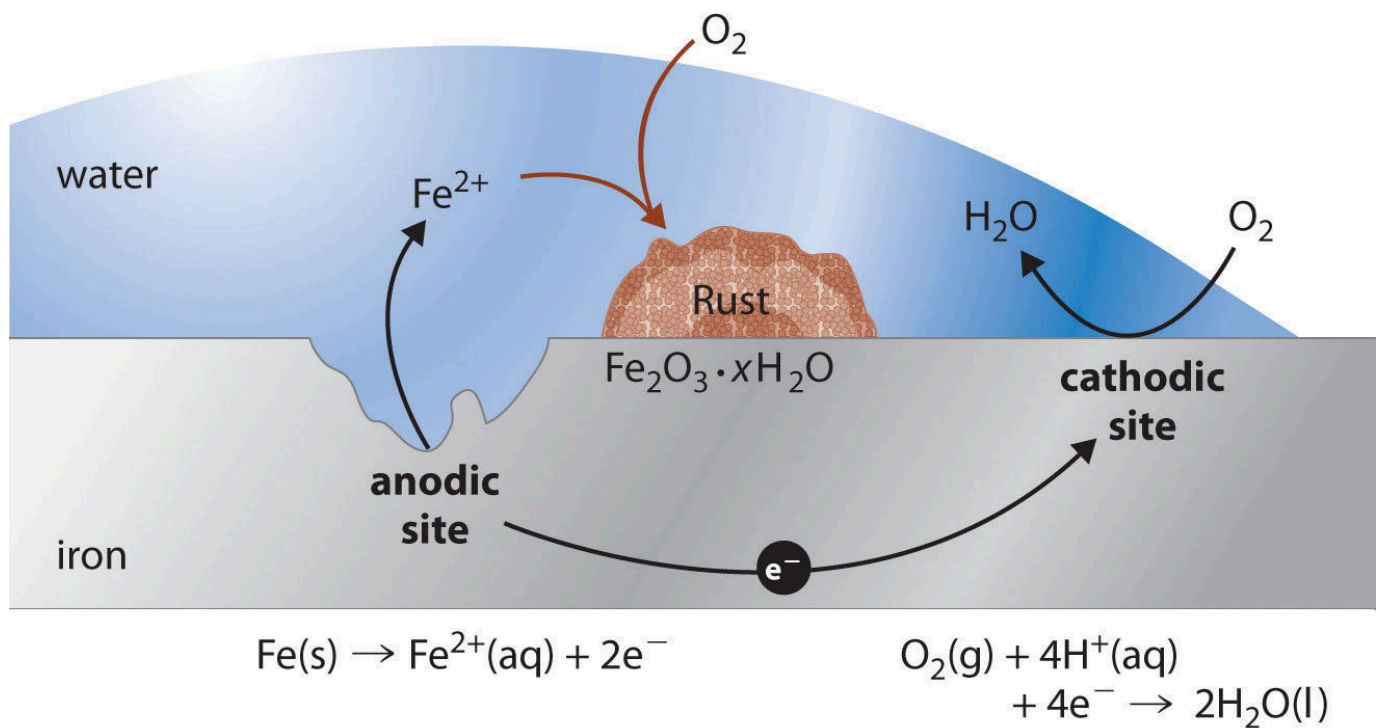
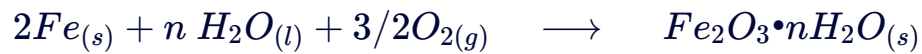
WIZE CONCEPT

You don't need to memorize the equation for the rechargeable lead-acid battery. Just know that rechargeable batteries have a discharging process (spontaneous, acts as a galvanic cell) and a charging process (non-spontaneous, requires a battery, acts as an electrolytic cell)!!

Corrosion and Prevention

Corrosion is the **oxidation (rusting)** of metals.

Iron rusts as follows:



- 1) Oxidation of Fe yields electrons that travel through the metal (anodic site)
- 2) Electrons that traveled through the metal reduce O_2 and form water (cathodic site)
- 3) Fe^{2+} migrates through the water drop and reacts with O_2 and H_2O to form rust

Corrosion prevention:

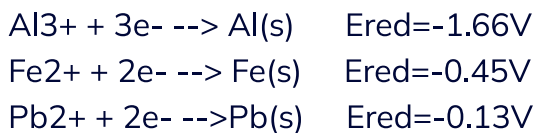
1. Coating

- **Galvanizing:** Zn Coating - this helps because Zinc is more easily oxidized than Fe is.
 - Zn acts as the anode instead of Fe!
 - If this is the case, how do the reduction potentials of Fe^{2+} and Zn^{2+} compare?
- Na_2CrO_4 should not be used to coat iron because it is toxic!

2. Cathodic protection (sacrificial anode)

- Instead of coating the iron with another metal, a **second metal is just placed in electrical contact with the first metal** and we call the second metal the "sacrificial anode"
 - Attach more active metals (e.g. Mg) to a material (e.g. iron tank/pipe) to be protected.
 - Mg is a better reducing agent than iron and will be oxidized first. The iron then becomes the cathode!

Given the following E_{red} , which would be more suitable to protect iron against corrosion, Al(s) or Pb(s)?



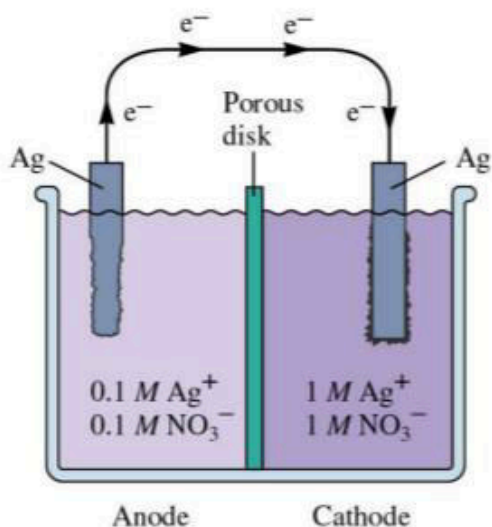
For this question we have to think about what happens to Fe during corrosion. During corrosion is Fe oxidized or reduced? _____

So if we are trying to protect against corrosion do we want Fe to be oxidized or reduced?

Now looking at the E_{red} , which metal would help to prevent corrosion? _____

Concentration Cells

Concentration Cells



- Consider the cell presented on the left.
- The $1/2$ cell reactions are the same, it is just the concentrations that differ.
- Will there be electron flow?

Are these standard or non-standard conditions? _____

If there is electron flow, when would the flow of electrons stop?

Problem:

There are two half-cells connected together. Both half-cells have a zinc electrode

and ZnCl_2 electrolyte. The concentration of Zn^{2+} in one cell is 0.20 M and in the other cell is more dilute but unknown. The voltage across the electrodes is 0.0289 V. Find the unknown $[\text{Zn}^{2+}]$.

We will need to use the Nernst equation since the conditions are _____ !

$$E_{\text{cell}} = E_{\text{ocell}} - (RT/nF)(\ln Q)$$

9.9 Electrolytic Cells

9.9.1

Electrolytic Cells

- Before we look at what an electrolytic cell looks like, **let's review what we already know about electrolytic cells:**
 - Do electrolytic cells have spontaneous or non-spontaneous electron flow?

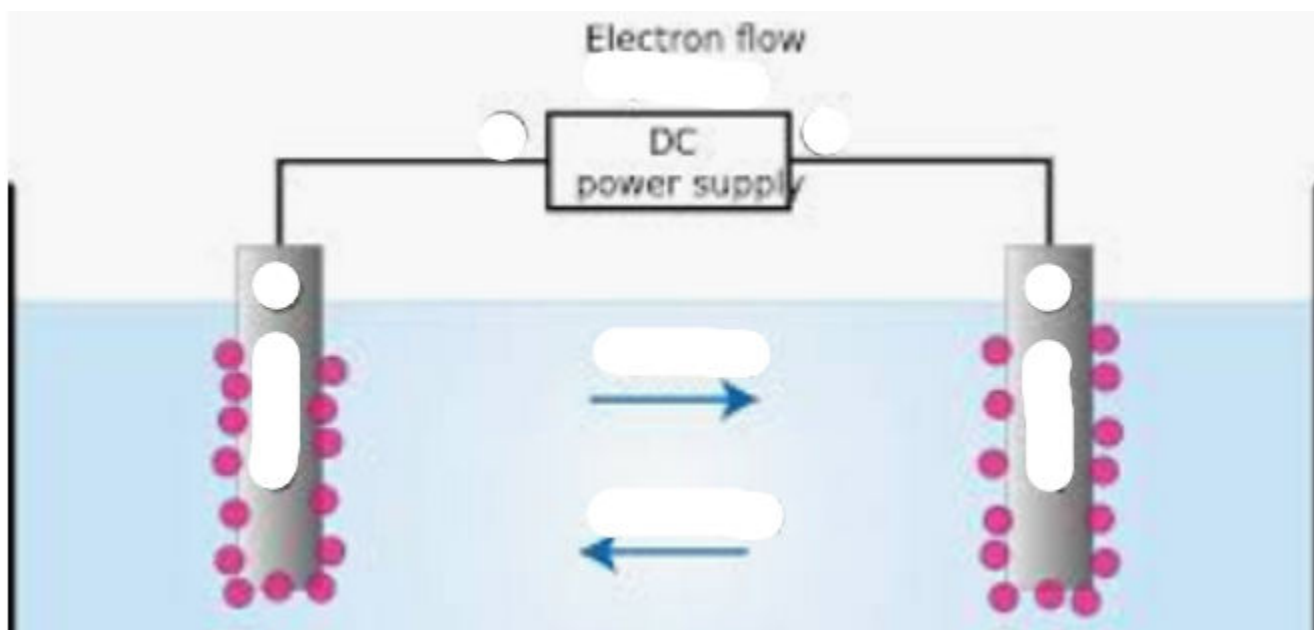
 - Is the ΔG° positive or negative? _____
 - According to the equation $\Delta G^\circ = -nFE^\circ_{\text{redox}}$ then is the E°_{redox} positive or negative?

Earlier we looked at a galvanic cell involving Zn(s) and Cu(s) electrodes.

We could also have an electrolytic cell with these same electrodes! Let's take a look at the half cells reactions from the table...



Which would be the oxidation reaction and which would be the reduction reaction if we want an electrolytic cell?



Based on this, label the following:

- 1) Let's say we are told that Zn(s) is on the left and Cu(s) is on the right. Label the anode and cathode.
- 2) Draw in electron flow.
- 3) Draw in charges on the electrodes and on the battery.

 WIZE CONCEPT

Note that because electrolytic cells have non-spontaneous flow of electrons, the only way to get the electrons to flow to where they don't want to go is by adding an external source of energy like a battery.

Electrons still flow from anode (where oxidation happens) to the cathode (where reduction happens).

But the difference between an electrolytic and galvanic cell is not only the battery needed for an electrolytic cell but also the charges on the electrodes.

In an electrolytic cell, electrons are going to where they don't want to go, but still going to the cathode. Therefore the cathode is negatively charged and the anode is positively charged!

- Also note that electrolytic cells do not require separation of the half reactions like galvanic cells do.

Faraday's Law of Electrolysis

There are 3 equations that go hand in hand for these types of problems:

$$1) Q=It$$

$$3) n=\frac{m}{M}$$

Q=charge (C)

n=moles

I=current (A)

m=mass (g) of metal

t=time (s)

M=molar mass (g/mol) of metal

$$2) Q=(n_{\text{of e-s}})(F)$$

n=moles of e-s involved in equation

F=Faraday's constant (96485C/mol e-)

Example: How long will it take to plate 1g of Au onto an Al ring if a current of 10A is applied?



Write out the variables we have and the ones we are looking for:

t=?

m=1g (Au)

I=10A

Solve for moles of Au:

$n=m/M$ m=1g, M=197g/mol

Find the moles of electrons:

Solve for Q:

$$Q = (n \text{ of } e's)(F)$$

Solve for time (t):

$$Q = It$$

Summary of Galvanic Vs Electrolytic Cell

Galvanic Cell:

- **Spontaneous** electron flow towards a positive charge
- Electrons go from the anode -> cathode where the **anode is negatively charged** and the **cathode is positively charged**

Electrolytic Cell:

- **Non-spontaneous** electron flow (requires an external source of energy like a battery)
- Electrons still go from the anode -> cathode but the **anode is positively charged** and the **cathode is negatively charged**

WATCH OUT!

If you see a cell being called an "electrochemical cell" this could be either a galvanic cell or an electrolytic cell.

We use the term "electrochemical cell" to refer to any cell.

Summary of Free Energy, Equilibrium, and Electromotive Force

$\Delta_r G$	E	Q	<i>Forward reaction spontaneity</i>
< 0	> 0	$< K$	<i>Spontaneous</i>
0	0	$= K$	<i>At equilibrium</i>
> 0	< 0	$> K$	<i>Non – spontaneous</i>

- Row 1 applies to a galvanic cell
 - Since the reaction is spontaneous in the forward direction, it wants to proceed towards the products and so $Q < K$ since the reaction is going to the right.
- Row 3 applies to an electrolytic cell
 - Since the forward reaction is non-spontaneous, that means the reverse reaction is spontaneous
 - This reaction would want to go from products to reactants
 - This reaction wants to go towards the left and has more products than reactants, which is why $Q > K$